FOREWORD

The Nigerian Educational Research and Development Council (NERDC) has the mandate to develop school curricula for all levels of the educational system in Nigeria. In line with Government adoption of the 9-Year Universal Basic Education (UBE) programme, NERDC, in 2006 developed a 9 – Year Basic Education Curriculum (BEC) to meet the ideals of the UBE programme. The curriculum accommodated the fundamentals of both the National Economic Empowerment and Development Strategies (NEEDS) and the Millennium Development Goals (MDG). The implementation of the curriculum commenced nationwide, in Primary 1 and JSS 1 classes in September 2008 respectively. The old curriculum was also being systematically phased out. The first batch of learners graduated in June 2010 after sitting the Basic Education Certificate Examination (BECE).

However, feedback received on the implementation of BEC called for an urgent need for the review of the 9-Year BEC to achieve a reduction in subject overload in the context of the number of subjects offered at the Primary and Junior Secondary Schools. We are pleased to note that the NERDC has revised the 9 – Year Basic Education Curriculum in line with global best practices and for the elimination of content overload and repetitions within and across subjects.

It is therefore for us, a great privilege to present the revised 9-Year Basic Education curriculum to all Nigerians for the use of our children now and in the future. I must congratulate all those who contributed to the review of this curriculum.

We wish to thank the Nigerian Educational Research and Development Council for keeping faith with its mandate, the High Level Policy Committee on Curriculum Development for preparing the policy grounds for the curricula, the stakeholders that made input into the revised curriculum framework and the resource persons for a job well done. It is our fervent hope that the teachers and learners for whom these curricula are produced would demonstrate commitment and assiduity in using these curricula. This is a proud legacy to leave for posterity.

With great expectations, we gladly recommend these curricula to all for the purpose of producing the best textual materials, the best in teaching performance, the best learning outcome, and most importantly, for attaining the goals we have set for ourselves in education in line with the Millennium Development Goals (MDGS) and in compliance with the National Economic Empowerment and Development Strategies (NEEDS).

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Honourable Minister of Education

CHIEF (BARR) EZENWO NYESOM WIKE
Honourable Minister of State for Education

PREFACE

Following the decision of the Federal Government to introduce the Universal Basic Education (UBE) programme, the Nigerian Educational Research and Development Council (NERDC) re-structured and re-aligned all extant primary and Junior Secondary Schools (JSS) curricula into a 9-Year Basic Education Curriculum for implementation in Nigerian schools. The 9-Year Basic Education Curriculum was particularly developed for the attainment of the Education for All (EFA) Goals, the critical targets of the National Economic Empowerment and Development Strategies (NEEDS), and the Millennium Development Goals (MDGs). Implementation of the 9-Year BEC commenced nationwide, in Primary 1 and JSS 1 classes in September 2008 respectively. The first batch of JSS students graduated in June 2011. It is expected that by September 2014, the cohort of pupils that benefitted from the use of BEC at the primary school level will be entering class one of the Junior Secondary School.

In October 2010, the President of the Federal Republic of Nigeria convened a National Stakeholders Forum to deliberate on the State of Education to in Nigeria. Delegates at the summit called for immediate action to reduce the number of subjects offered at the Basic Education level to between 6 and 13 subjects. Consequently, NERDC was directed to review the 9-Year BEC in line with the recommendations of the Summit, reflecting at every point global best practices and contemporary national concerns.

The curriculum revision process involved consultations with stakeholders (curriculum experts, subject matter specialists, teachers, policy makers, employers of labour, parents, etc.) in education at various levels (Concept Formulation, High Level Policy Committee (HLPC), etc.) to prepare a **Conceptual Framework** for the review of the 9-Year BEC. The Framework identifies and groups related disciplines, thereby achieving a reduction in subject listings. For example, related UBE subjects curricula like Home Economics, Agriculture are brought together to create a new UBE subject curriculum to be called **Pre-Vocational Studies.** Similarly, Islamic Studies, Christian Religious Studies, Social Studies, Civic Education, etc. that focus primarily on the inculcation of values (societal, moral, interpersonal) now form a new UBE subject called **Religious and Value Education**. Key concepts in the former curricula will now form integrating threads for organizing the contents of the new subject into a coherent whole.

Thus the Conceptual Framework for the review of BEC comprises of a ten (10) subjects namely:

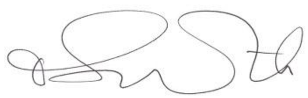
1. English Studies
2. Mathematics
3. Basic Science and Technology
4. Religion and National Values
5. Cultural and Creative Arts
6. Business Studies

7. Nigerian Languages
8. Pre-vocational Studies
9. French
10. Arabic

In the process of the review, particular efforts have been made to further reduce the overload within and across subjects without compromising depth, appropriateness, and interrelatedness of the curricula contents. Specifically, the revised 9-Year BEC address amongst other things, the issues of value re-orientation, poverty eradication, peace and dialogue, including human rights education, family life/HIV and AIDS education, critical thinking, entrepreneurship and life skills as well as encourage innovative teaching and learning approaches and techniques. In addition, the curriculum is organized to ensure continuity and flow of themes, topics and experiences from primary school to junior secondary school levels.

Since the curriculum represents the total experiences to which all learners must be exposed, the contents, performance objectives, activities for both teachers and learners, teaching and learning materials and evaluation guide are provided. The prescriptions represent the minimum content to be taught in the schools in order to achieve the objectives of the 9-year basic education programme. However, teachers are encouraged to enrich the contents with relevant materials and information from their immediate environment, but adapting the curriculum to their needs and aspirations. Thus the curriculum can be adapted for such special needs as nomadic education, non-formal education and education of the physically challenged.

In conclusion, I wish to express profound gratitude to the President of the Federal Republic of Nigeria, Dr. Goodluck Ebele Jonathan, for his vision and commitment to improved education delivery and the Senior Special Assistant to the President on MDGs for the financial support of received during the review, printing and implementation of the revised 9-year Basic Education Curriculum. I congratulate and commend the Concept Development, Planning, Writing, Critique and Editorial teams which were drawn from all parts of this country for an excellent job; as well as acknowledge the efforts of the Policy and Policy Programmes Unit of the Executive Secretary's Office, the Curriculum Development Centre of the NERDC that coordinated the various activities and workshops that led to the successful review of the 9-Year Basic Education Curriculum within a short period of time. It is our belief that when these curricula are properly implemented, the future generations of this great country will be better for it.



Prof. Godswill Obioma fman, fiica, fcon, fnae

Executive

INTRODUCTION

This curriculum is the outcome of the restructuring and realignment of the 9 – Year Basic Education Mathematics Curriculum for Primary and Junior Secondary School (JSS). The revision became necessary because:

1. the first batch of JSS students under the Universal Basic Education (UBE) programme graduated in June 2011;
2. of the feedback from the implementation of the curricula in schools that identified repetition and duplication of concepts.
3. recommendations of the Presidential Summit on Education (2010) suggested the reduction in the number of subjects offered in Primary and Junior Secondary Schools;
4. of the need to encourage innovative teaching and learning approaches and techniques that promotes creativity and critical thinking in learners.
5. of the need to infuse emergent issues that are of national and global concern such as gender sensitivity globalization and entrepreneurship.

In this review, some topics were dropped while new ones were added. Also, there are shifts in topics from one class to the other where necessary.

THE OBJECTIVES OF Mathematics CURRICULUM

The 9-Year Basic Education Mathematics Curriculum (Revised: 2012) is focused on giving learners the opportunity to::

- acquire mathematical literacy necessary to function in an information age.
- cultivate the understanding and application of mathematics skills and concepts necessary to thrive in the ever changing technological world.
- develop the essential element of problem solving, communication, reasoning and connection within the study of mathematics
- take advantage of the numerous career opportunities provided by mathematics;
- become prepared for further studies in mathematics and other related fields.

STRUCTURE OF THE CURRICULUM

The thematic approach was also adopted selecting the content and learning experiences in the curriculum. The thematic approach to curriculum organisation is useful for accommodating emerging issues without necessarily disrupting the curriculum structure. The Themes and Sub-Themes in the9-Year BEC (revised, 2012) are:

THEME	SUB-THEME
Number and Numeration	**Whole Number** Fraction
Basic Operations	Basic Operations Derived Function Derived Operations
Algebraic Processes	Algebraic operations Open Sentences
Mensuration and Geometry	Primary Measures Secondary Measures Shapes
Every Day Statistics	Data Collection and Presentation Chance and Events

It should also be noted that this revised curriculum placed emphasis on affective domain and quantitative reasoning unlike the previous curriculum. This is to boost learners' cognitive and psychomotor capabilities.

The 9-Year Basic Education Mathematics Curriculum is a teaching curriculum. Consequently, it provides maximal aid for the teacher by prescribing topics, objectives or expected learning outcomes, pupils and teaches activities and evaluation guides. However, for this curriculum to be effective in achieving the purpose for which it is meant:

1. it is strongly recommended that copies of these documents be made available to every primary school teacher and much emphasis placed on its use in order to achieved stated objectives.
2. there is need to organize workshops for teachers supervisors and inspectors on how to interpret and use the curriculum.
3. the minimum qualification for teachers teaching in Basic 1 to Basic 9 is National Certificate in Education (NCE), University graduates of first degree and above in various disciplines teaching in Basics 1 – 9 must have education qualification for the Basic Education Programme to succeed.

Finally, it is our hope that the revised Mathematic Curriculum will achieve the goals and objectives of the Universal Basic Education (UBE) Programme in Nigeria as contained in the National Policy on Education (2004) and the UBE bill of 2005.

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MATHEMATICS CURRICULUM